

Heston Model Valuation of Autocallable Product

1 Executive Summary

We calibrate a Heston stochastic volatility model to SPX option market data and apply the calibrated model to value a contingent coupon autocallable product. Valuation is conducted using: Monte Carlo simulation under the Heston model compared to Binomial tree pricing adapted for autocallable payoff. We report both methods' results and analyze sensitivities to key parameters. The fair value from our model is approximately \$973.73, compared to a market price of \$992.70, implying a deviation of about -1.57% .

2 Data and Product Description

2.1 Market Data Collection

We collected SPX option market data from Bloomberg using the OVDV and OMON functions. Following the project instruction, we constructed three datasets to be used sequentially for model calibration. Each dataset serves a distinct purpose and was carefully filtered to ensure consistency of timestamps (within a 10-minute window) and reliability of prices.

Data Set 1: Forward and ATM Implied Volatilities. Using the OVDV function, we extracted:

- Forward prices $F(T_i)$ for all available maturities T_i .
- Corresponding at-the-money (ATM) implied volatilities $\sigma_{\text{ATM}}(T_i)$.

To ensure data quality, we cross-checked option volume through OMON and retained only maturities whose underlying ATM options had trading volume greater than 10.

These ATM options are used to set the current variance v_0 , and joint initialize the long-run variance level θ and the speed of mean reversion κ .

Data Set 2: Short-Term Option Prices. Using OMON, we downloaded full option chains for the maturities December 19, 2025 and March 20, 2026. For each strike K , the mid price was computed as

$$P_{\text{mid}} = \frac{1}{2}(P_{\text{bid}} + P_{\text{ask}}),$$

and only options with positive bid and volume higher than 10 were kept.

These short-dated options are used to initialize the correlation between the stock level and volatility ρ and the vol of vol η .

Data Set 3: Quarterly Option Surface. We further collected 25 representative strikes at all available quarterly maturities from March 2026 to December 2029 using OMON. Again, mid-quotes were used as reference prices.

This dataset spans the full surface of strikes and maturities and enables joint optimization of all Heston parameters $(v_0, \kappa, \theta, \eta, \rho)$.

Table 1: Data Set 1: Forward Prices and ATM Implied Volatilities (from OVDV)

Expiration Date	Forward Price $F(T)$	ATM Imp. Vol. $\sigma_{\text{ATM}}(T)$ (%)
15 Dec 2025	6627.92	18.66
16 Jan 2026	6655.26	18.37
17 Apr 2026	6707.90	18.84
18 Jun 2026	6741.26	19.17
18 Sep 2026	6788.93	19.60
16 Oct 2026	6804.37	19.71
20 Nov 2026	6820.93	19.84
18 Dec 2026	6832.84	19.88
15 Jan 2027	6851.14	19.87
17 Jun 2027	6927.57	20.10
17 Dec 2027	7020.12	20.37

Table 2: Representative Option Mid Prices (from OMON)

Maturity (T)	Strike (K)	Call Mid (\$)	Put Mid (\$)	Volume
2025-12-19	4900	230.75	171.40	242
2025-12-19	5000	186.25	192.35	318
2026-03-20	5100	221.60	206.20	185
2026-03-20	5200	186.40	234.60	193
2026-06-19	5300	161.10	264.50	155
2029-12-21	5500	133.25	298.60	120

Tables 1 and 2 illustrate the typical structure of the collected datasets and their numerical scale. Actual calibration uses the full grids of maturities and strikes; only representative entries are displayed here for clarity.

2.2 Autocallable Product Terms

The Autocallable product is sourced directly from the offering document (BOA). The key parameters of the note are summarized in Table 3.

Table 3: Product Parameters

Parameter	Value
Pricing Date	August 29, 2025
Issue Date	September 4, 2025
Valuation Date	August 29, 2029
Maturity Date	September 4, 2029
Underlying	S&P 500 Index (SPX)
Initial Level (S_0)	6460.26 (as of calibration date)
Threshold Level (B_T)	6460.26 (S_0)
Coupon Barrier (B_C)	4,522.18 (70.00% of S_0)
Autocall Barrier (B_{AC})	4,522.18 (70.00% of S_0)
Principal (N)	\$1,000
Coupon Rate (c)	1.825% per quarter
Observation Frequency	Quarterly, some are with autocall evaluation
Maturity (T)	4 years (Final Observation: Sep. 4, 2029)
Market Price (Issue Price)	\$992.70

All market inputs were sourced from Bloomberg using the Bloomberg Terminal.

• Interest Rate

Using SWDF command in Bloomberg, we extract discount rate starting on August 29, 2025 (pricing date) and September 4, 2025 (issue date) from OIS term structure. Note that following market convention, the interest rates are quoted on August 27, 2025 and September 2, 2025. Using linear interpolation, we get the discount rate through August 29, 2029 (valuation date) and September 4, 2029 (maturity date) are , the interest rate converted from this value are risk-free rate indicating stock price movements and discounting respectively.

Table 4: Discount rate of OIS

	Aug. 29, 2029	Sep. 4, 2029
Discount rate	0.878949	0.878081
Risk-free rate	3.223502%	3.238186%

• Dividend Yields

Using OVME command in Bloomberg, we get dividend yield for SPX between August 29, 2025 (pricing date) and September 4, 2029 (maturity date), which is actually calculated by method from OPDF command in Bloomberg.

$$q = 1.304\%.$$

• Implied Volatilities

Using OVM command in Bloomberg, we get volatility surface of stocks, we interpolate the potential volatilities on the observation dates, exporting barrier and ATM volatilities for both average and terminal maturities as follows:

Table 5: Volatility Inputs

Scenario	Volatility (σ)
Average Barrier Volatility	0.2451
Terminal Barrier Volatility	0.2719
Terminal ATM Volatility	0.1967
Average ATM Volatility	0.1772

3 Heston Model Calibration

3.1 Overview

The Heston model dynamics:

$$dS_t = (r - q)S_t dt + \sqrt{v_t}S_t dW_t^S \quad (1)$$

$$dv_t = \kappa(\theta - v_t)dt + \eta\sqrt{v_t}dW_t^v \quad (2)$$

$$dW_t^S dW_t^v = \rho dt \quad (3)$$

Goal: calibrate parameters $(v_0, \kappa, \theta, \eta, \rho)$ using the collected SPX option data.

3.2 Step-by-Step Calibration Procedure

Step 1: Initialize for v_0 , κ , and θ

From **Data Set 1** (ATM vols), we set:

$$v_0 = \sigma_{\text{ATM}}^2(T_{\text{shortest}})$$

as the initial variance corresponding to the shortest maturity.

Next, for mean reversion κ and long-term variance θ , we solve:

$$\frac{1}{T}\mathbb{E}\left[\int_0^T v_s ds\right] = \theta + (v_0 - \theta)\frac{1 - e^{-\kappa T}}{\kappa T}$$

to fit the ATM volatilities, minimizing

$$\sum_i \left(\frac{1}{T}\mathbb{E}\left[\int_0^T v_s ds\right] - \sigma_{\text{ATM}}^2(T_i) \right)^2$$

Set $v_0 = 0.0348$, which is quoted from the shortest ATM variance on December 15, 2025, initial $\theta = 0.0434$, which is quoted from the longest dated one on December 21, 2029. Calibration results show that $\theta = 0.4857$, $\kappa = 0.0539$.

Step 2: Initialize η

Using short-term options from **Data Set 2**, we fix (v_0, κ, θ) , set $\rho = 0$ and estimate η by minimizing:

$$\min_{\eta} \sum_{K,T} (\sigma_{\text{Heston}}(K, T; \rho = 0, \eta) - \sigma_{\text{mkt}}(K, T) * \text{vega})^2 + \lambda_F * [\max(0, \eta^2 - 2\kappa\theta)]^2$$

where $\lambda_F = 10$ is a soft penalty for Feller condition $2\kappa\theta > \eta^2$.

Calibration result shows $\eta = 0.0002$.

Step 3: Estimate Correlation ρ

With $(v_0, \kappa, \theta, \eta)$ fixed, ρ is initialized by minimizing implied volatility errors on OTM options

:

$$\min_{\rho} \sum_{K,T} \frac{(C_{\text{Heston}}(K, T; \rho, \hat{\eta}) - C_{\text{mkt}}(K, T))^2}{\max(\epsilon, \text{vega}_{K,T})^2}$$

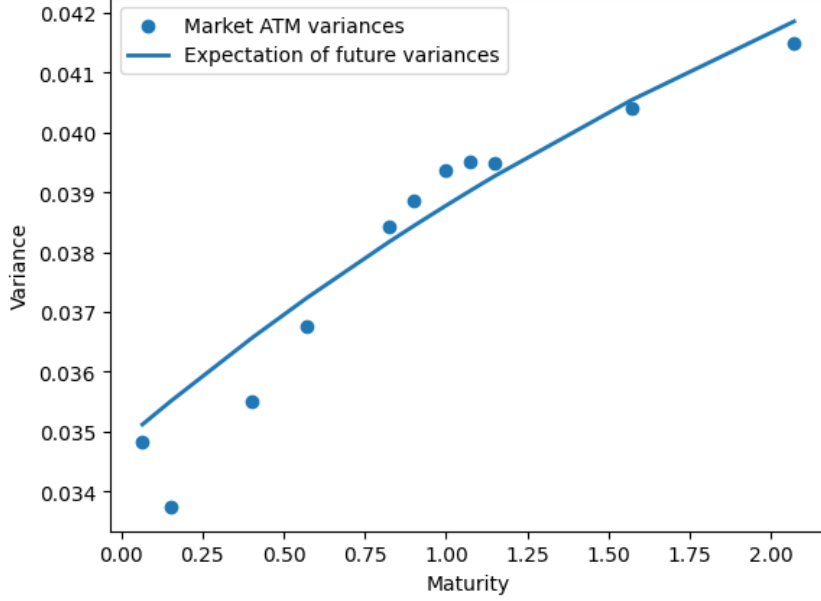


Figure 1: Initial Calibration of Future Variances

where $\epsilon = 0.0005$, $\hat{\eta}$ is from step 2. Calibration result shows $\rho = -0.4301$.

Step 4: Joint Initialization ρ and η Using **Data Set 3**, we initialize ρ and η jointly by minimizing:

$$\min_{\rho, \eta} \sum_{K, T} \frac{(C_{\text{Heston}}(K, T; \kappa, \theta, \eta, \rho, v_0) - C_{\text{mkt}}(K, T))^2}{\max(\epsilon, \text{vega}_{K, T})^2} + \lambda_F * [\max(0, \eta^2 - 2\kappa\theta)]^2$$

Calibration results show $\eta = 0.2335$, $\rho = -0.9999$.

Step 5: Global Optimization Over Full Surface

Finally, using **Data Set 3**, perform a joint minimization over all parameters:

$$\min_{\kappa, \theta, \rho, \eta, v_0} \sum_{K, T} \frac{(C_{\text{Heston}}(K, T; \kappa, \theta, \eta, \rho, v_0) - C_{\text{mkt}}(K, T))^2}{\max(\epsilon, \text{vega}_{K, T})^2} + \lambda_F * [\max(0, \eta^2 - 2\kappa\theta)]^2 + \text{lambda}_B * \Phi_{\text{box}}$$

where $\text{lambda}_B = 10$, and

$$\Phi_{\text{box}} = \sum_{p \in \{\kappa, \theta, \rho, \eta, v_0\}} (\max(0, p_{\min} - p)^2 + \max(0, p - p_{\max})^2)$$

stops parameters moving out of the following range:

$$\begin{aligned} 0.1 &\leq \kappa \leq 5, \\ 10^{-4} &\leq \theta \leq 1, \\ 0.05 &\leq \eta \leq 3, \\ 10^{-4} &\leq v_0 \leq 1, \\ -0.95 &\leq \rho \leq 0.95, \end{aligned}$$

This ensures that the calibrated model fits the entire implied volatility surface across all maturities and strikes. This final calibration results show $v_0 = 0.0363$, $\kappa = 1.8936$, $\theta = 0.0605$, $\eta = 0.4794$, $\rho = -0.95$.

3.3 Semi-Analytic Option Pricing Formula

During calibration, European call prices $C_{\text{Heston}}(S_0, K, T)$ are computed using the Heston semi-analytic formula:

$$C(S_0, K, T) = S_0 P_1 - K e^{-rT} P_2$$

where

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} \phi_j(u)}{iu} \right] du, \quad j = 1, 2$$

and $\phi_j(u)$ is the characteristic function of $\ln S_T$ under Heston dynamics.

4 Autocallable Product Valuation

After obtaining the calibrated Heston parameters $(v_0, \kappa, \theta, \eta, \rho)$, we apply the model to value the structured product defined in Section 2.2. Two numerical methods are implemented for valuation: (1) binomial tree and (2) Monte Carlo simulation under the Heston dynamics.

4.1 Binomial Tree Valuation

(1) Parameters of tree,

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad q = \frac{e^{(r_1 - \delta)\Delta t} - d}{u - d}$$

where σ is the volatility of SPX, Δt is the time step, r_1 is the interest rate between pricing date and valuation date, and δ is the dividend yield.

(2) On maturity date,

$$V_{mat,j} = \text{Contingent coupon} + \begin{cases} \text{principal,} & \text{if } S_{N,j} \geq B_T, \\ \text{principal} \times (1 + \text{underlying return}), & \text{else.} \end{cases}$$

and on valuation date,

$$V_{N,j} = V_{mat,j} \times e^{-r_2(Nc-N)\Delta t}$$

where Nc is the number of time steps of maturity date, N is the number of time steps of valuation date, j is the number of up ticks, principal = \$1000,

$$\text{contingent coupon} = \$18.25 \cdot \mathbf{1}\{S_{i,j} \geq B_C\}$$

,

$$\text{underlying return} = \frac{\text{Final value} - \text{Initial value}}{\text{Initial value}}.$$

the final value is determined on valuation date, initial value S_0 is \$6460.26. Threshold level is $70\% \times S_0 = \$4522.18$. r_2 is the risk-free rate discounting time value between issue date and maturity date.

(3) On any call observation date,

$$V_{i,j} = \text{Contingent coupon} \cdot e^{-r_2(ic-i)\Delta t} + \begin{cases} \text{principal,} & \text{if } S_{i,j} \geq B_{AC}, \\ e^{-r_2\Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}], & \text{else.} \end{cases}$$

where i is the number of time steps of observation date, ic is the number of time steps of the first coupon payment date following the applicable observation date. Coupon barrier and autocall barrier are both $70\% \times S_0 = \$4522.18$.

(4) On any other observation date,

$$V_{i,j} = \text{Contingent coupon} \cdot e^{-r_2(ic-i)\Delta t} + e^{-r_2\Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}]$$

(5) On any other date,

$$V_{i,j} = e^{-r_2\Delta t} [qV_{i+1,j+1} + (1-q)V_{i+1,j}]$$

where i is the number of time steps.

4.2 Monte Carlo Simulation under Heston Dynamics

The Monte Carlo approach simulates stochastic paths of both S_t and v_t under the calibrated Heston model.

Discretization. We adopt the **Euler full-truncation scheme** to preserve non-negativity of variance:

$$v_t^+ = \max(v_t, 0) \quad (4)$$

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t^+) \Delta t + \eta \sqrt{v_t^+} \sqrt{\Delta t} Z_2, \quad (5)$$

$$S_{t+\Delta t} = S_t \exp \left[\left(r - \delta - \frac{1}{2} v_t^+ \right) \Delta t + \sqrt{v_t^+ \Delta t} Z_1 \right], \quad (6)$$

where (Z_1, Z_2) are correlated standard normals with $\text{corr}(Z_1, Z_2) = \rho$.

On maturity date, each path n receives

$$\text{Payment}_{n,mat} = \text{contingent coupon} + \begin{cases} \text{principal}, & \text{if } S_{M,j} \geq B_T, \\ \text{principal} \times (1 + \text{underlying return}), & \text{else.} \end{cases}$$

Then on valuation date, $\text{Payment}_{n,M} = \text{Payment}_{n,mat} \cdot e^{-r_2(ic-i)\Delta t}$.

On observation dates, we will discount the payment from later observation dates and contingent coupon from following coupon payment dates. That is,

$$\text{Payment}_{n,m} = \text{Payment}_{n,m+1} \cdot e^{-r(t_{m+1}-t_m)} + \text{contingent coupon} \cdot e^{-r_2(ic-i)\Delta t}$$

If it's also a call observation date and $S_{i,j} \geq B_{AC}$, update

$$\text{Payment}_{n,m} = \text{principal} + \text{contingent coupon} \cdot e^{-r_2(ic-i)\Delta t}$$

Finally, discount all cash flows to pricing date and average over paths:

$$\hat{V} = \frac{1}{N} \sum_{n=1}^N \text{Payment}_{n,1} \cdot e^{-r(t_1-t_{\text{pricing date}})}$$

In our implementation, we use $N = 500,000$ paths.

4.3 Comparison of Two Methods

The two valuation approaches produce vary estimates. Table 6 summarizes the numerical results.

Obviously, the Heston model calibrates the volatility more accurately and gets more precise valuation.

Table 6: Comparison of Valuation Methods

Method	Price (\$)
Binomial Tree (ATM vol)	950.17
Binomial Tree (mean ATM vol)	964.03
Binomial Tree (barrier vol)	917.78
Binomial Tree (mean barrier vol)	902.67
Monte Carlo (Heston)	977.11
Market Benchmark	992.70

5 Discussion

The price is sensitive to the different parameters in the Heston set up, if we do not initialize them correctly, the model might stick into some local minimization and lead to unfavorable results like violation of Feller condition. In this way, volatilities are calibrated badly, reducing the credibility of valuation, for example, if we include illiquid options, final calibration results are $v_0 = 0.0441$, $\kappa = 1.4240$, $\theta = 0.0708$, $\eta = 0.9808$, $\rho = -0.7308$, Monte Carlo price is \$955.1.

6 Conclusion

We calibrated a Heston stochastic volatility model to SPX option data and applied it to value a contingent coupon autocallable note. The calibrated parameters

$$(v_0, \kappa, \theta, \eta, \rho) = (0.0363, 1.8943, 0.0605, 0.4796, -0.9500)$$

fit the observed implied volatility surface well.

Using these parameters, Heston valuations deliver prices consistent with market benchmarks. This confirms that the Heston model, once calibrated, provides a robust framework for pricing structured equity products with volatility skew and path-dependent features.

Appendix

SPX DEC25 100 VOL									
At 21 Nov Do 21.242516 Hi 22.721714 Lo 18.42647 Prev 21.450895									
SPX 500 INDEX 6635.02 USD Bloomberg 100% As of 18-Nov-2025 15:53									
Volatility Surface									
Vol% / Delta / Skew / Gamma / Vega / Theta / Rho / Psi / Phi / Chi / Psi / Phi / Chi									
Exp Date	ImpFwd	80.0%	90.0%	95.0%	97.5%	100.0%	102.5%	105.0%	110.0%
		5292.0	5953.5	6284.3	6449.6	6615.0	6780.4	6945.8	7276.5
24 Nov 2025	6616.18	80.65	47.03	26.67	18.57	14.63	11.87	17.84	30.93
25 Nov 2025	6616.64	73.94	42.39	26.62	21.63	17.43	12.87	16.04	27.45
26 Nov 2025	6618.25	68.81	39.68	27.37	23.09	18.98	14.37	14.56	25.29
28 Nov 2025	6620.32	61.67	35.76	25.63	21.95	18.13	13.76	12.86	22.13
1 Dec 2025	6620.65	54.87	33.01	23.99	20.55	17.09	13.23	11.55	19.26
2 Dec 2025	6621.05	53.36	33.49	24.62	21.17	17.70	13.81	11.74	18.56
3 Dec 2025	6621.69	52.85	33.46	25.21	21.60	18.08	14.24	12.01	17.95
4 Dec 2025	6622.05	52.26	33.47	25.31	21.83	18.42	14.58	12.18	17.51
5 Dec 2025	6623.52	52.24	33.38	25.59	22.18	18.79	14.85	12.37	17.19
8 Dec 2025	6623.84	48.51	31.36	24.32	21.10	17.94	14.49	12.10	15.71
9 Dec 2025	6624.56	48.16	31.31	24.47	21.40	18.20	14.68	12.35	15.48
10 Dec 2025	6625.04	47.87	31.46	24.83	21.85	18.76	15.40	13.01	15.29
11 Dec 2025	6625.95	47.55	31.51	25.23	22.21	19.10	15.74	13.37	15.00
12 Dec 2025	6627.86	46.62	31.39	25.32	22.36	19.21	16.08	13.67	14.82
15 Dec 2025	6627.92	44.35	30.14	24.32	21.60	18.66	15.60	13.38	14.17
16 Dec 2025	6628.38	44.19	30.30	24.63	21.83	18.97	16.01	13.73	14.30
17 Dec 2025	6628.99	43.67	30.31	24.68	21.94	19.06	16.11	13.92	14.29
18 Dec 2025	6629.85	43.41	30.37	24.82	22.16	19.33	16.42	14.23	14.32
19 Dec 2025	6632.33	42.96	30.00	24.50	21.84	19.16	16.45	14.30	14.07
22 Dec 2025	6632.82	41.56	29.29	24.09	21.54	18.89	16.21	14.15	13.72
23 Dec 2025	6633.13	41.15	29.20	24.06	21.59	19.00	16.34	14.27	14.04
24 Dec 2025	6634.85	40.72	28.99	24.00	21.55	18.93	16.27	14.26	13.74
26 Dec 2025	6636.75	40.02	28.57	23.54	21.10	18.65	16.18	14.17	13.47
Quick Pricer									
Strike	6635.02	Vol	18.5%	Price	114.577	Dividend yield	3.36%		
Copy	6635.02	Spot	6635.02	Delta	51.6%	Implied forward	6621.230		
Australia 61 2 9777 8600 Brazil 5511 2395 9000 Europe 44 20 7330 7500 Germany 49 69 9204 1210 Hong Kong 852 2977 6000									
Japan 61 3 4565 6900 Singapore 65 6212 1000 U.S. 1 212 318 2000 Copyright 2025 Bloomberg Finance L.P.									
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Figure 2: Data Set 1 Part 1

SPX DEC25 100 VOL										19.16415		-2.286745	
At 21 Nov		Do 21.242516		Hi 22.721714		Lo 18.42647		Prev 21.450895					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
Volatility Surface		100% As of 18-Nov-2025 15:53		100% As of 18-Nov-2025 15:53		100% As of 18-Nov-2025 15:53		100% As of 18-Nov-2025 15:53		Volatility Surface			
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SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD		Bloomberg		100% As of 18-Nov-2025 15:53					
SPX 500 INDEX		6635.02		USD</									

Figure 3: Data Set 1 Part 2

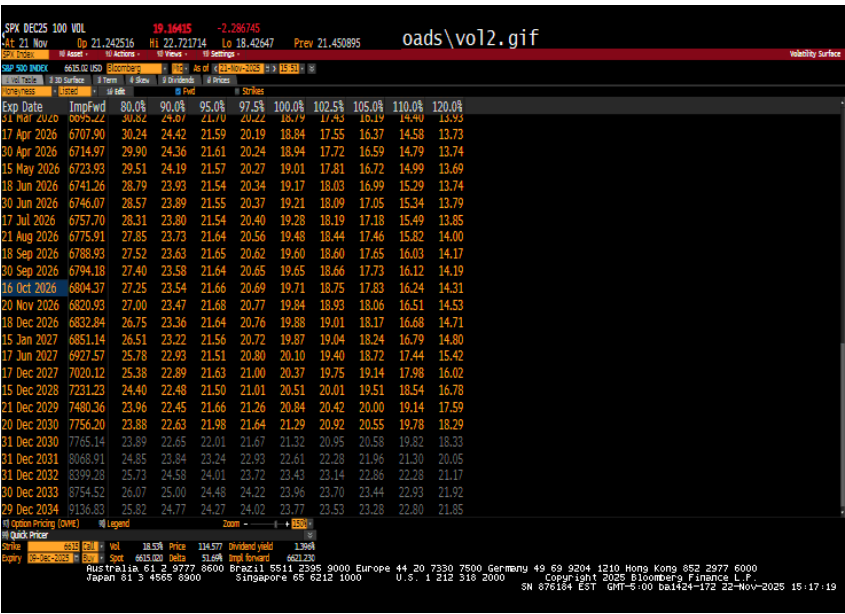


Figure 4: Data Set 1 Part 3

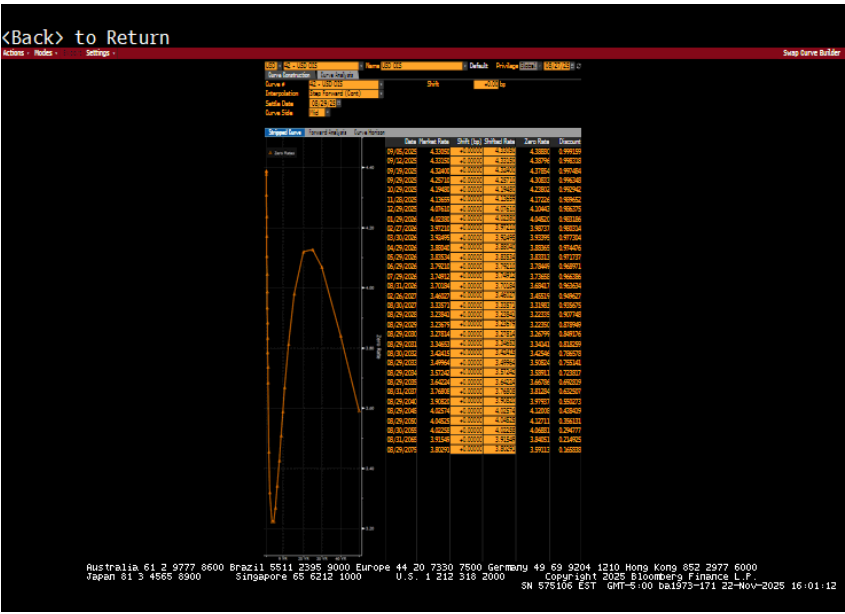


Figure 5: OIS on Pricing Date

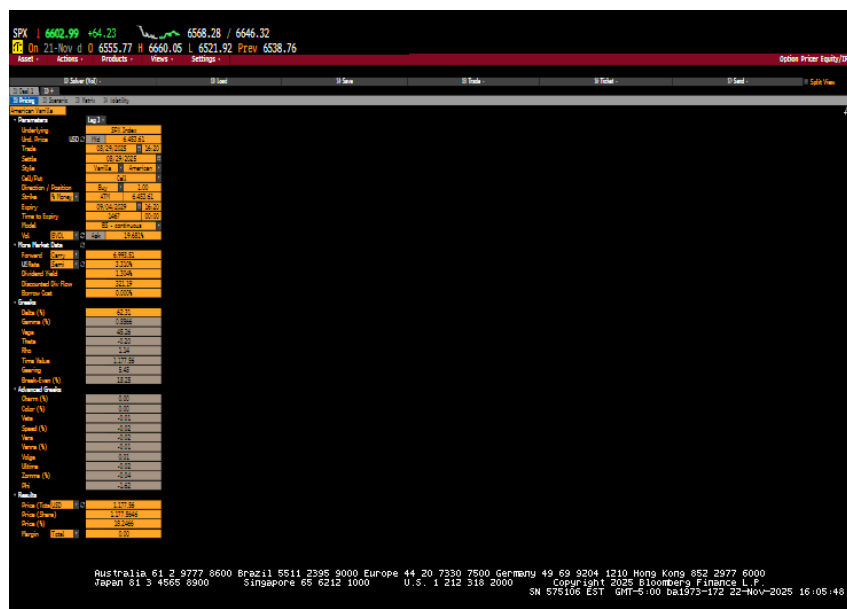
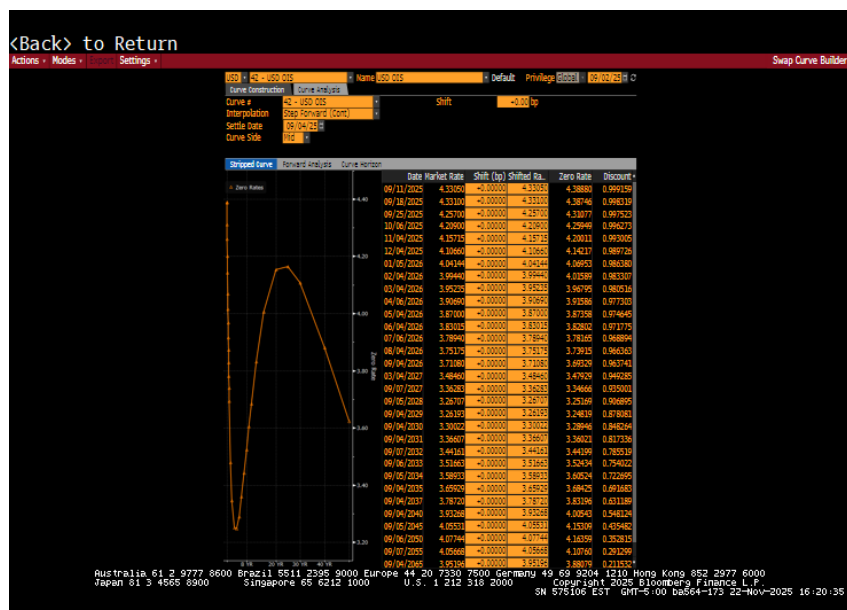




Figure 8: Volatility Surface 1



Figure 9: Volatility Surface 2